logdist effect of a covariate

Parametrization

This model implements a non-linear effect of a positive covariate x as a part of the linear predictor,

$$\beta\,(1 + \exp(\alpha_1 \log(x) - \alpha_2 x)$$

where $\beta \in \Re$, $\alpha_1, \alpha_2 \in \Re^+$ and $x \geq 0$.

Hyperparameters

This model has three hyperparameters, the scaling β, α_1 and α_2.

$$\theta_1 = \beta \qquad \theta_2 = \log(\alpha_1) \qquad \theta_3 = \log(\alpha_2)$$

and the priors are given for θ_1, θ_2 and θ_3.

Specification

```
f(x, model="logdist", hyper = ..., precision = <precision>)
```

where `precision` is the precision for the tiny noise used to implement this as a latent model.

Hyperparameter specification and default values

doc `A nonlinear model of a covariate`

hyper

theta1

hyperid `39021`
name `beta`
short.name `b`
initial `1`
fixed `FALSE`
prior `normal`
param `0 1`
to.theta `function(x) x`
from.theta `function(x) x`

theta2

hyperid `39022`
name `alpha1`
short.name `a1`
initial `0`
fixed `FALSE`
prior `loggamma`
param `0.1 1`
to.theta `function(x) log(x)`
from.theta `function(x) exp(x)`

theta3

hyperid 39023
name alpha2
short.name a2
initial 0
fixed FALSE
prior loggamma
param 0.1 1
to.theta function(x) log(x)
from.theta function(x) exp(x)

constr FALSE

nrow.ncol FALSE

augmented FALSE

aug.factor 1

aug.constr

n.div.by

n.required FALSE

set.default.values FALSE

pdf logdist

Example

```
logdist = function(x, beta,  alpha)
{
    return (beta * (1 + exp(alpha[1] * log(x) - alpha[2] * x)))
}

n = 1000
s=0.1
x = runif(n)
beta = 1
alpha = c(1, 0.5)
## start at the true values
hyper = list(
    beta = list(initial = beta),
    a1 = list(initial = log(alpha[1])),
    a2 = list(initial = log(alpha[2])))
## start somewhere else
hyper = list(
    beta = list(initial = 1),
    a1 = list(initial = 0),
    a2 = list(initial = 0))

y = logdist(x, beta,  alpha) + rnorm(n, sd = s)
r = (inla(y ~ -1 + f(x, model="logdist", hyper = hyper),
```

```
          data = data.frame(y, x),
          family = "gaussian",
          verbose=TRUE,
          control.inla = list(h=0.0001),
          control.family = list(
              hyper = list(
                  prec = list(
                      initial = log(1/s^2),
                      fixed = TRUE)))))
summary(r)
```

Notes

None